

1.1 Coastal Landscapes

Geography — Eduqas A-Level (A110QS) — Component 1: Changing Landscapes

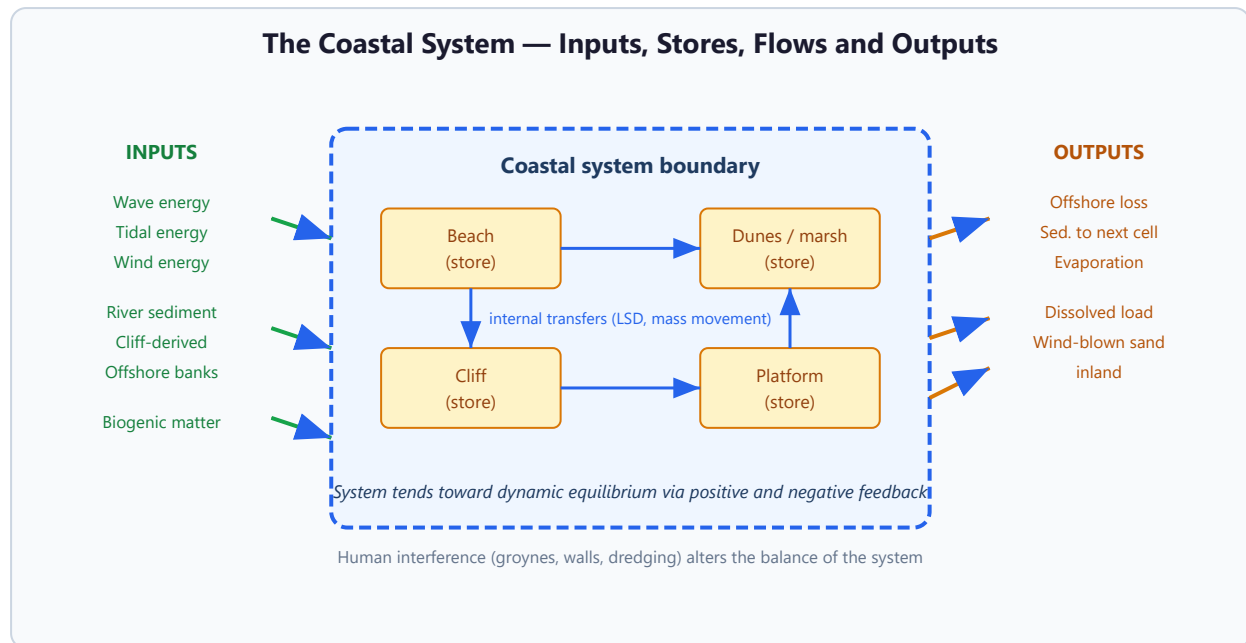
1. The Coast as a System

The Eduqas specification requires students to adopt a *systems approach* when explaining coastal landscapes. The coast is treated as an **open system** because both energy (from waves, tides and wind) and matter (sediment, dissolved minerals, water) cross its boundaries. Coastal landscapes are the product of interactions between four sub-systems: the *marine* (waves, tides, currents), *sub-aerial* (weathering, mass movement), *terrestrial* (rivers supplying sediment, geology) and *atmospheric* (wind, temperature, precipitation). Human activity is increasingly regarded as a fifth controlling factor.

Coastal system: An open system at the interface of land, sea and atmosphere in which energy and sediment are exchanged through inputs, stores, transfers/flows and outputs. It tends towards a *dynamic equilibrium* — a balance of inputs and outputs that fluctuates around a long-term average.

System components you must be able to label and explain:

- **Inputs** — marine energy (wave, tidal, current), solar and wind energy, sediment from rivers, cliff erosion, offshore banks and biogenic sources (shells, coral), precipitation.
- **Stores** — beaches, spits, bars, tombolos, sand dunes, salt marshes, cliffs, wave-cut platforms, offshore sand banks.
- **Transfers / flows** — longshore drift, swash and backwash, aeolian (wind) transport, mass movement (rockfall, slumping, sliding), sub-aerial weathering processes.
- **Outputs** — sediment exported offshore or to neighbouring cells, evaporation, dissolved load carried in solution, landform erosion products carried inland or lost to the sea floor.



Dynamic equilibrium: A state in which a system oscillates around a long-term average. Short-term disturbance (e.g. a single winter storm stripping sand) is offset by recovery (calm-weather constructive waves rebuilding the beach). The system is dynamic because it constantly changes, but equilibrium because the mean state is preserved.

Feedback loops are vital A-level content. *Negative feedback* stabilises the system: cliff collapse adds sediment to a beach, the wider beach dissipates wave energy through friction, and further erosion rates slow. *Positive feedback* amplifies change: storm removal of beach sand exposes the cliff base, storm waves attack the cliff directly, producing more erosion and narrowing the beach still further. The recognition that human intervention can *flip* feedback polarity — e.g. sea-wall construction removing a beach sediment source and accelerating downdrift erosion — is a key AO3 evaluation point.

Exam tip — systems language: Eduqas examiners reward the correct use of system vocabulary (inputs, stores, transfers, outputs, feedback, equilibrium). In 15-mark extended responses, open with a systems definition, classify the landform or process within the system, and return to the concept of equilibrium or feedback in your conclusion. Generic “the beach is formed by waves” sentences lose AO1 marks.

2. Sources of Energy and Sediment

Wave Energy

Waves are generated when wind drags the ocean surface. The size of a wave and its potential energy depend on three variables:

- **Fetch** — distance of open water over which the wind has blown.
- **Wind speed** — higher wind speed transfers more kinetic energy to the sea surface.
- **Wind duration** — the longer the wind blows, the larger the wave can grow.

Bathymetry (the shape of the seabed) further modifies wave energy as it reaches shallow water: a steeply shelving bed produces *plunging* destructive waves, while a gently shelving shore produces *spilling* constructive waves.

Fetch: The uninterrupted distance of open water across which wind blows in a single direction. The UK's south-west coast faces the Atlantic with a fetch of roughly 5,000 km, generating large prevailing swell. East coasts have a shorter fetch from the North Sea but experience destructive north-easterly storm waves when polar lows track south.

Sediment Sources

Sediment entering a coastal system derives from a number of sources, each contributing to the *sediment budget*:

- **Cliff erosion** — the dominant source along high-energy, soft-rock coasts (e.g. Holderness supplies ~1.8 million m³ of sediment per year to the southern North Sea).
- **Rivers** — globally the largest input, though dam construction has reduced delivery to many coasts.
- **Offshore transport** — constructive waves and tidal currents move sand and gravel from offshore banks onto the beach.
- **Longshore drift from adjacent cells** — some sediment crosses cell boundaries at low-angle transitions.
- **Biogenic sources** — shells, coral fragments and calcareous algae. Important in tropical coasts and parts of western Scotland.
- **Aeolian and glacial inputs** — wind-blown sand from interior deserts (minor in the UK) and relict glacial sediment (very significant in eastern England).

Sediment cell (littoral cell): A largely self-contained stretch of coastline bounded by prominent physical features (usually large headlands) within which the inputs, transfers and outputs of sediment form a closed system. England and Wales are divided into 11 major sediment cells. These form the management units for Shoreline Management Plans (SMPs).

Sediment budget: The balance between sediment inputs and outputs within a cell. A *positive* budget produces progradation (coast builds outward); a *negative* budget produces retreat. Human interference (groynes trapping drift, dam construction starving rivers of bedload, offshore dredging) can switch a coast from positive to negative budget within decades.

3. Marine Processes

Wave Types: Constructive and Destructive

Two distinct wave regimes alternate along most temperate coasts. *Constructive* waves dominate in calmer summer months and tend to build up beaches; *destructive* waves dominate in winter storms and strip sediment offshore. The net profile of a beach is therefore seasonal.

Feature	Constructive Waves	Destructive Waves
Wave height	Low (<1 m typically)	High (>1 m, often 3–5 m in storms)
Wavelength	Long (up to 100 m)	Short (<20 m)
Frequency	Low (6–8 per minute)	High (10–14 per minute)
Swash vs backwash	Strong swash > weak backwash	Weak swash < strong backwash
Net effect on beach	Sediment moves onshore; berms, ridges build up	Sediment pulled offshore; steep scarp develops
Breaker type	Spilling breakers	Plunging breakers

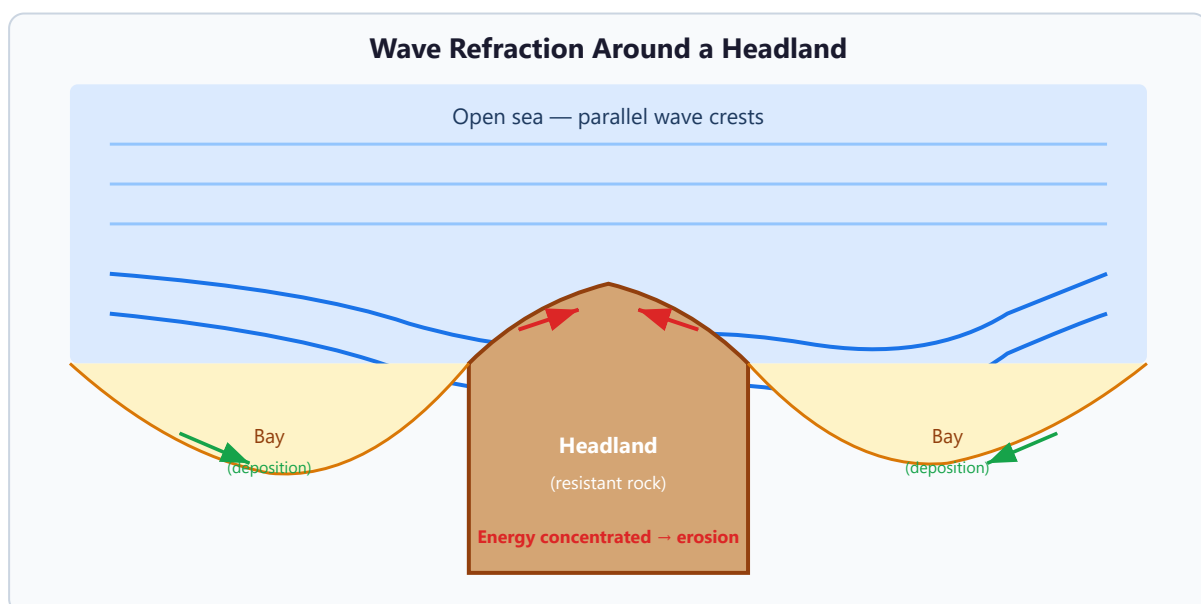
Typical conditions	Settled weather; long-period swell from distant storms	Locally generated wind seas in stormy conditions
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Constructive wave: A low-energy wave (height < 1 m), long wavelength and low frequency, in which a powerful swash carries sediment up the beach while a weak backwash leaves most of it deposited. Result: a gently sloping, wide beach with pronounced berms.

Destructive wave: A high-energy wave (height > 1 m), short wavelength and high frequency, whose weak swash deposits little, but whose strong backwash drags sediment seaward. Result: a steep, narrow beach profile and offshore bar formation.

Wave Refraction

Waves arriving at an irregular coast of bays and headlands are refracted. In shallow water around a headland they slow through seabed friction, while the part of the wave crest still over deep water advances faster. The wave crest bends so that its energy converges on the headland. In bays the opposite occurs: wave crests diverge and energy is dispersed over a wider frontage, encouraging deposition. Over geological time, refraction progressively straightens a coast. Where waves strike a steep cliff or sea wall they may instead be **reflected** back out to sea with little loss of energy; reflected waves interfere with incoming ones and can scour the base of the structure.



Tides

Tides are generated primarily by the gravitational pull of the Moon, modulated by the Sun. The difference between high and low water (tidal range) controls the vertical extent of the intertidal zone where marine processes can operate. The Severn Estuary has one of the largest tidal ranges in the world (~14 m on spring tides), producing wide mudflats and salt marshes — classic features the Eduqas spec expects candidates to know. Tides also drive offshore and onshore tidal currents, which are strongest in narrow straits and estuary mouths and can move sand and silt independently of waves. Most UK coasts have two high and two low tides a day (semi-diurnal); some coasts elsewhere have only one (diurnal).

Coastal Erosion Processes

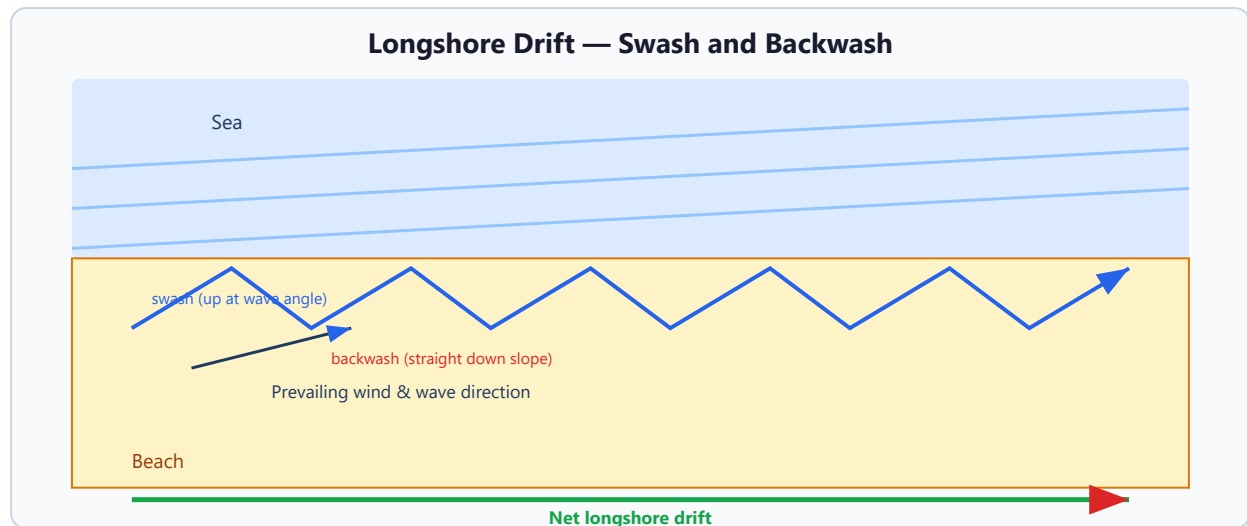
- **Hydraulic action** — the sheer force of breaking waves compressing air into cracks in the cliff face. When the wave retreats, the pressurised air explodes outward, prising rock apart.
- **Abrasion (corrasion)** — pebbles and sand carried in the waves are hurled against the cliff, acting like sandpaper. The most effective cliff-toe erosion process on rocky coasts.
- **Attrition** — sediment particles collide and wear each other down, becoming progressively smaller and more rounded. This modifies the sediment itself, not the cliff directly.
- **Solution (corrosion)** — chemical dissolution of soluble minerals, especially calcium carbonate. Limestone coasts (e.g. the Jurassic Coast around Lulworth) are particularly vulnerable.

Rates of cliff retreat depend on *rock lithology* (soft boulder clay retreats at metres per year; granite at millimetres), *geological structure* (jointing, bedding, dip direction), *wave energy*, *cliff profile* and the presence or absence of a protective beach.

Sediment Transport

- **Traction** — rolling of pebbles and boulders along the seabed.
- **Saltation** — bouncing of sand grains in short hops.
- **Suspension** — finer silt and clay held within the turbulent water column.
- **Solution** — dissolved load transported invisibly.

Longshore drift (LSD): The net along-shore movement of sediment, driven by waves arriving at an oblique angle to the coast. The *swash* carries sediment up the beach at the angle of wave approach; *backwash* returns it down the steepest gradient (perpendicular to the waterline) under gravity. The resulting zig-zag transfer can move sediment tens of kilometres along a coast.



Exam tip — LSD precision: A common C-grade mistake is writing “waves push the sediment along the beach”. For Eduqas A-level you must specify: (1) waves approach at an angle because of the prevailing wind, (2) swash moves sediment up the beach at that angle, (3) backwash returns it perpendicular to the shoreline under gravity, (4) repetition produces net zig-zag transport. Always name the process and the agent.

Deposition

Deposition occurs wherever wave energy is reduced below the threshold required to keep sediment in transport. Common settings include:

- Sheltered embayments where wave refraction disperses energy.
- Estuaries where tidal currents meet riverine outflow and velocity falls.
- Behind obstacles such as headlands, islands, or man-made breakwaters.
- At the end of spits where coastlines change orientation and wave trains no longer carry sediment forward.
- **Flocculation** — in estuaries, tiny clay particles carry electrical charges; when river water mixes with salt water they clump together into larger, heavier flocs that settle out, building mudflats. As energy falls, sediment is also **sorted** by size, the coarsest settling first.

4. Sub-aerial Processes

Sub-aerial processes operate on the *land* side of the cliff and include weathering and mass movement. They prepare sediment for marine removal and, in the case of mass movement, can themselves be dramatic landscape-forming events.

Mechanical (Physical) Weathering

- **Freeze-thaw (frost shattering)** — water penetrates cracks, freezes (expanding by ~9 %), widens the crack and eventually prises rock apart. Most active where temperatures oscillate around 0 °C.
- **Salt weathering** — seawater evaporating in cracks leaves salt crystals whose growth exerts pressure on the surrounding rock, flaking it apart. Important in spray zones of warm coasts.
- **Wetting and drying** — repeated hydration and dehydration causes expansion and contraction, especially in clays.

Chemical Weathering

- **Carbonation** — slightly acidic rainwater (H_2CO_3) reacts with limestone to form calcium bicarbonate, which is soluble and washed away.
- **Solution** — direct dissolution of soluble minerals (halite, gypsum).
- **Oxidation** — iron-bearing minerals oxidise in air and water, weakening rock structure and producing the reddish-brown staining seen on many sandstone cliffs.
- **Hydrolysis** — feldspars in granite react with water to form clay minerals, gradually disintegrating the granite.

Biological Weathering

- Plant roots wedging open joints as they grow.
- Burrowing animals (crabs, molluscs, piddocks) drilling into soft rock.
- Organic acids from lichens and algae weakening rock surfaces.

Mass Movement

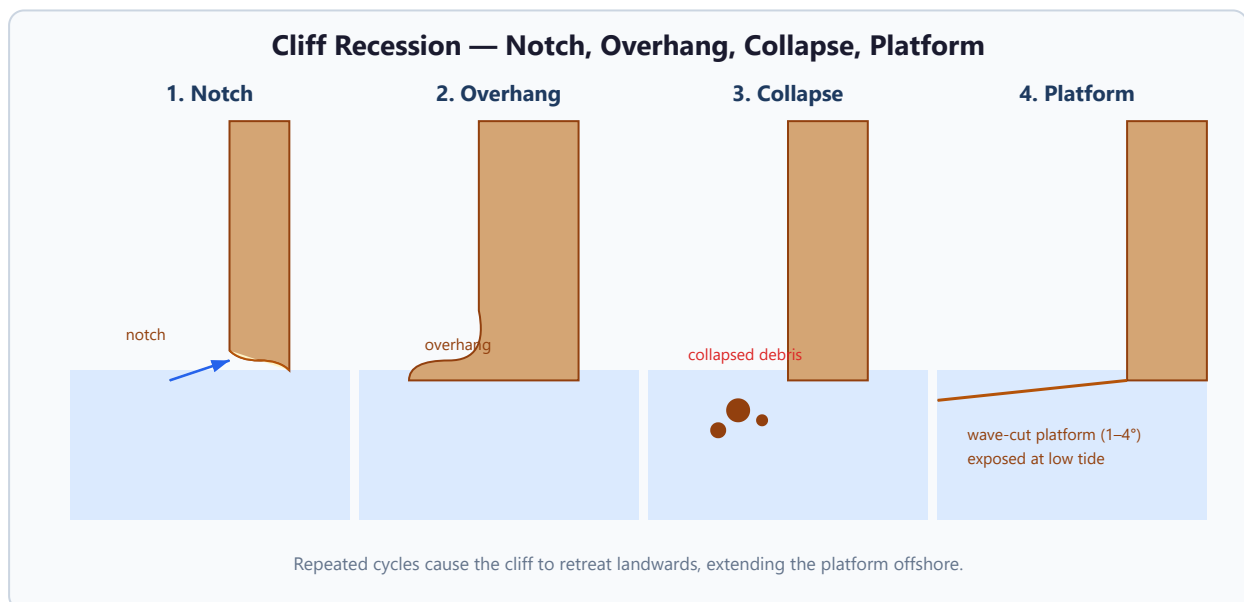
Mass movement: The downslope movement of rock, regolith and soil under gravity. Includes rockfall, slides, slumps and flows. It becomes dominant in preparing cliff faces for marine removal wherever the cliff is lubricated by water, unsupported at its base, or composed of weak or layered material.

- **Rockfall** — sudden collapse of loose blocks from a vertical or near-vertical face, typical of jointed limestone or chalk cliffs (e.g. Beachy Head collapses).
- **Rotational slip / slump** — along a curved failure surface; common in softer cliffs like the Holderness boulder clay. The cliff face rotates backwards at the top, producing a stepped (“terraced”) profile.
- **Translational slide** — movement along a planar surface, often a bedding plane or fault.
- **Flow** — water-saturated regolith moving as a viscous fluid, ranging from slow earth flows to rapid mudflows.

5. Erosional Landforms

Cliffs and Wave-Cut Platforms

A cliff is formed where waves attack a steep section of coast. Hydraulic action and abrasion concentrate at the high-tide line, excavating a **wave-cut notch**. Continued undercutting produces an unsupported overhang which eventually collapses under gravity, and the cliff retreats landwards. The eroded base is left as a gently sloping (1–4°) **wave-cut platform** exposed at low tide, often deeply pitted with rock pools and striations.



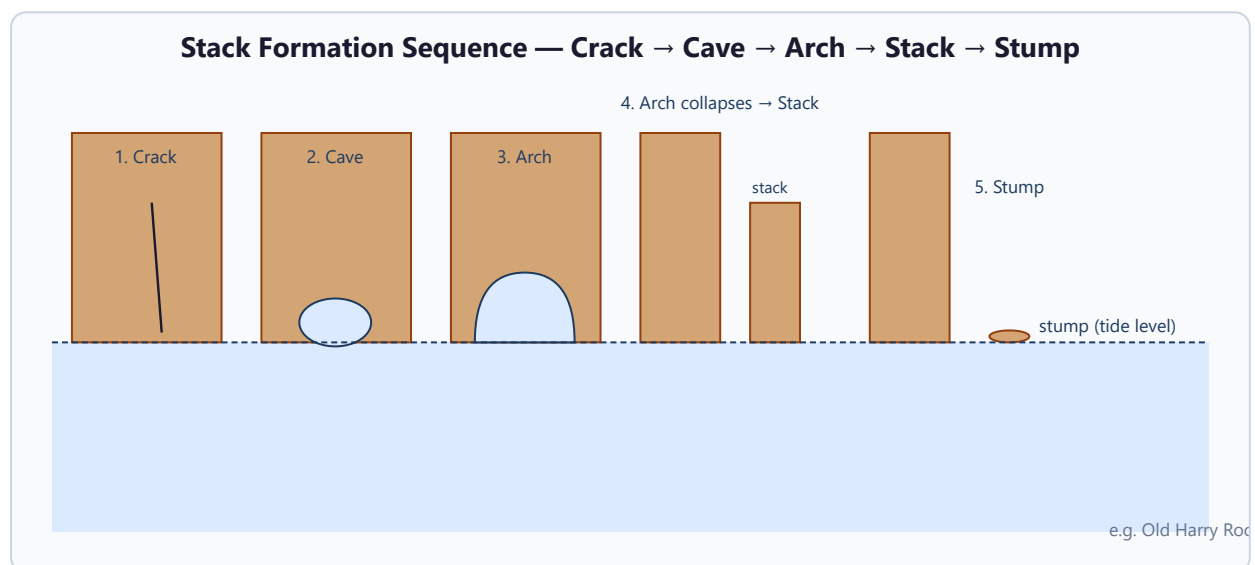
Headlands and Bays

Differential erosion of alternating rock outcrops produces headlands and bays. Two contrasting scenarios appear in exam questions:

- **Discordant coastline** — bands of rock run perpendicular to the coast. Less resistant rock is eroded faster, producing bays between resistant headlands. Classic example: the Dorset coast around Studland and Swanage, with chalk headland at Ballard Point and bay cut into softer sands.
- **Concordant coastline** — bands of rock run parallel to the coast. The more resistant outer band acts as a natural wall; once breached, the softer rock behind is rapidly eroded to form a cove. Classic example: Lulworth Cove, Dorset.

Caves, Arches, Stacks and Stumps

Where a resistant headland contains lines of weakness (faults, joints), waves exploit these to produce a characteristic erosional sequence. This is one of the most predictable exam landform questions, and examiners look for *named landforms* used in the correct order.



1. **Crack** — hydraulic action exploits a joint or fault in the headland.
2. **Cave** — the crack is enlarged by abrasion and hydraulic action.
3. **Arch** — the cave is cut right through the headland, leaving a natural bridge.
4. **Stack** — the arch roof becomes unsupported and collapses, leaving a pillar of rock.
5. **Stump** — further erosion reduces the stack to a low rock visible only at low tide.

Geos and blowholes: A *geo* is a long, narrow, steep-sided inlet cut along a line of weakness. A *blowhole* forms when the roof of a sea cave is breached by air pressure, producing a vertical shaft that “blows” spray upwards in storms (e.g. the Bullers of Buchan, Scotland).

6. Depositional Landforms

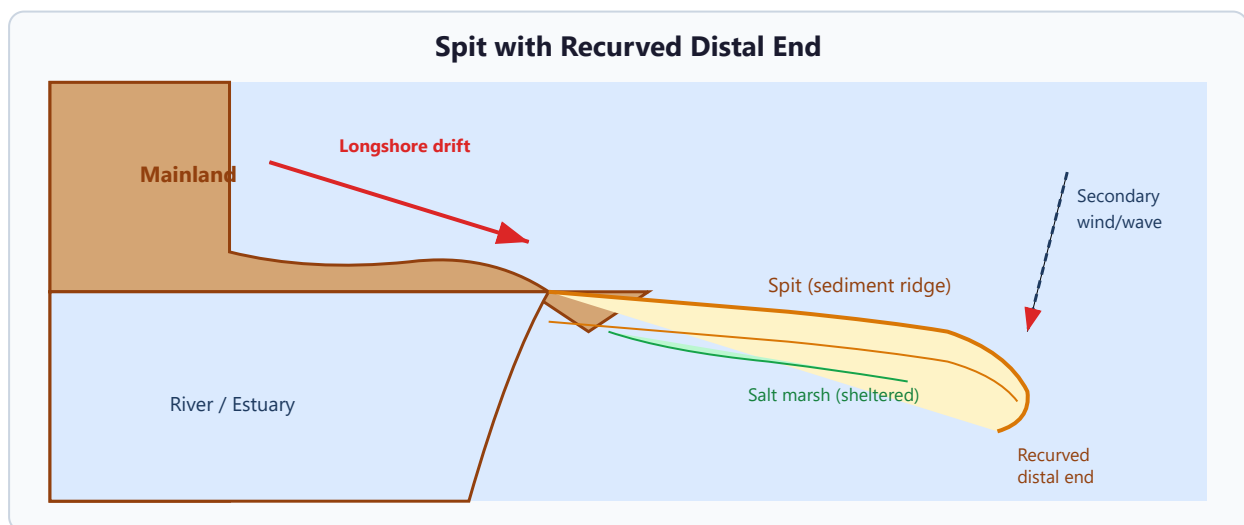
Beaches

Beaches are accumulations of sediment between the low-water mark and the highest storm-tide line. Their morphology is controlled by the balance of constructive and destructive waves, sediment supply and grain size.

- **Swash-aligned beaches** — waves approach parallel to the shore; sediment moves shore-normally, producing broadly symmetrical profiles (e.g. pocket beaches in bays).
- **Drift-aligned beaches** — waves approach at a consistent angle, producing active longshore drift along their length (e.g. Holderness beaches).
- **Berms** — ridges of sediment marking the high-tide line of successive tides.
- **Cusps** — crescentic depressions on the foreshore, formed by a self-organising interaction of swash and backwash.
- **Storm ridges** — coarser material thrown to the back of the beach by exceptional storms.

Spits

A spit is a long, narrow ridge of sand or shingle attached to the land at its proximal end and terminating in open water at its distal end. It forms where longshore drift carries sediment past an indentation in the coastline (e.g. an estuary mouth). Sediment is deposited beyond the turn as water depth increases and waves lose competence. The distal end often curves inland (“recurve”) as secondary winds or waves from a different direction push it shorewards. Salt marshes develop in the sheltered water behind.



UK examples: *Spurn Point* (Holdingness), ~5 km long, periodically breached in storms; *Hurst Castle Spit* (Solent), shingle spit with multiple recurve ridges recording former positions.

Bars

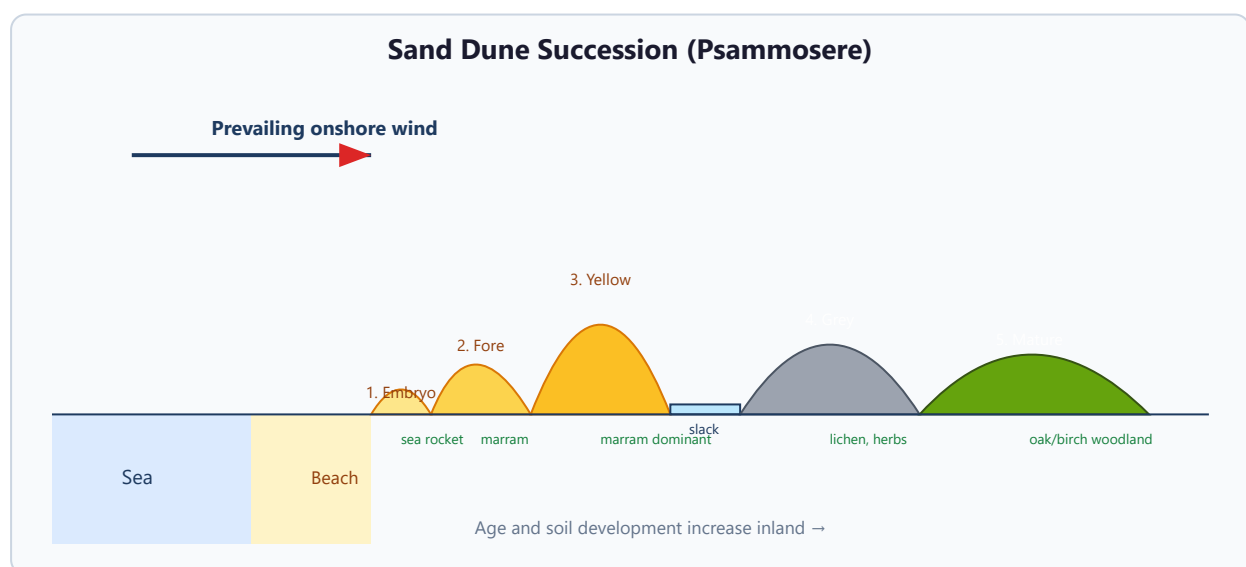
A bar forms where a spit extends across the mouth of a bay, connecting two headlands and trapping a body of water behind it as a *lagoon*. Conditions required include a gently shelving offshore profile and no large river to keep the gap open. Slapton Ley (south Devon) is a classic UK bar-lagoon system with freshwater lagoon behind a shingle bar.

Tombolos

A tombolo is a bar connecting an offshore island to the mainland, formed where wave refraction around the island produces a zone of deposition in its lee. The UK's outstanding example is Chesil Beach, a 29 km shingle tombolo linking the Isle of Portland to the Dorset mainland, with the Fleet lagoon enclosed behind. A **cusate foreland** is a triangular accumulation of shingle projecting from the coast, built where longshore drift from two directions meets (for example, where waves arrive alternately from the south-west and east). Dungeness in Kent is the UK's classic example.

Sand Dunes

Sand dunes form on wide, sandy beaches where the prevailing wind blows onshore across a large drying intertidal area. Vegetation succession progressively stabilises the dunes, producing a characteristic cross-section from shoreline to climax community.



The pioneer species, *sea rocket* and *saltwort*, tolerate salt spray and trap wind-blown sand. Once sand accumulates and salt exposure decreases, *marram grass* (*Ammophila arenaria*) colonises, with deep roots that bind sand and rhizomes that spread laterally. Yellow dunes are high and unstable; grey dunes are older, stabilised and have enough humus to support herbs and lichens. Mature/climax dunes far inland may support pine or oak woodland. *UK examples*: Branton Burrows (Devon), Newborough Warren (Anglesey) — common Eduqas fieldwork sites.

Salt Marshes

Salt marshes develop in sheltered estuaries and behind spits, where halophytic (salt-tolerant) plants trap suspended sediment at high tide. Zonation runs from *mudflat* (bare, algae only) through *pioneer zone* (*Salicornia*, cord-grass *Spartina*), *low marsh* (sea aster, sea lavender) to *high marsh* (saltmarsh grass, rushes) and finally climax community above the highest tide line. Vertical accretion rates typically $1\text{--}5\text{ mm yr}^{-1}$ — broadly in step with current sea-level rise, which makes salt marshes important natural coastal defences. Outside the UK, **biotic processes** build two further coastal landscapes: **coral reefs** (fringing, barrier and atoll forms), built by reef-building corals in warm (roughly $18\text{--}30\text{ }^{\circ}\text{C}$), clear, shallow, well-lit tropical water, and **mangrove coasts**, where salt-tolerant trees with dense aerial roots trap fine sediment on sheltered tropical shores (see the Sundarbans case study below).

7. Sea-Level Change

Sea level is never static on geological timescales. Two mechanisms operate:

Eustatic change: A global change in sea level caused by a change in the volume of water in the oceans. Primarily driven by the growth or melting of ice sheets (during glacial maxima, sea level fell $\sim 120\text{ m}$ below present) and by thermal expansion of seawater with warming.

Isostatic change: A local change in the vertical position of the land, producing a *relative* change in sea level. After the last glaciation, the removal of ice has allowed northern Britain to rebound upwards ("glacio-isostatic rebound"), so that in Scotland relative sea level has *fallen*, whereas in southern England the crust is still subsiding and relative sea level has risen.

Feature	Eustatic	Isostatic
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Scale	Global	Local / regional
Mechanism	Ocean volume change	Crustal vertical movement
Timescale	Fast (decades–millennia)	Slow (millennia)
Direction in UK	Currently rising	N rising; S sinking
Driver example	Ice-sheet melt, thermal expansion	Glacial unloading, sediment loading

Emergent Landforms (relative sea-level fall)

- **Raised beaches** — former beach platforms now perched above present high-tide level; classic examples on the west coast of Scotland, including the Isle of Arran.
- **Fossil (relict) cliffs** — steep slopes inland of the current coast, marking a former active cliff line.
- **Emergent marine platforms** — wave-cut platforms lifted above their zone of formation.

Submergent Landforms (relative sea-level rise)

- **Rias** — drowned river valleys; broad, shallow, V-shaped in cross-section. Classic examples on the south-west peninsula: Fal Estuary, Dartmouth, Kingsbridge.
- **Fjords** — drowned glacial valleys; deeper, U-shaped cross-section, often with a shallow rock bar at the mouth (the glacier's terminal zone). Norway, western Scotland, Alaska, Chile, New Zealand. Loch Torridon is a UK fjord.
- **Dalmatian coasts** — drowned concordant coasts producing long, parallel island chains offshore; classic example Croatia's Adriatic coast, from which the type name derives.

8. Modern and Future Sea-Level Rise

Since the industrial revolution, global mean sea level has risen by ~20 cm, with the rate accelerating from ~1.5 mm yr⁻¹ in the 20th century to ~4 mm yr⁻¹ in the early 21st century. The IPCC Sixth Assessment Report (AR6, 2021) projects further rise by 2100 of:

- **SSP1-2.6** (low emissions) — 0.32–0.62 m
- **SSP2-4.5** (intermediate) — 0.44–0.76 m

- **SSP5-8.5** (high emissions) — 0.63–1.01 m, with a low-likelihood-high-impact upper bound approaching 2 m if marine ice-sheet instability operates.

Two physical processes contribute: *thermal expansion* (~50 % of rise to 2100) and *mass addition* from melting glaciers and the Greenland and Antarctic ice sheets. Consequences include:

- Greater frequency and severity of storm surge flooding as smaller storms can breach defences designed for lower baseline sea level.
- “Coastal squeeze” where hard defences are fixed and rising seas cannot retreat landwards, destroying intertidal habitat.
- Accelerated cliff retreat on soft-rock coasts.
- Salinisation of groundwater in low-lying coasts (e.g. Ganges–Brahmaputra delta).
- Existential threat to small island developing states (Tuvalu, Kiribati, Maldives) where mean elevation is <2 m.

Exam tip — use up-to-date data: For the evaluative sections of a 15-mark extended response, quote specific IPCC AR6 figures (e.g. “0.44–0.76 m by 2100 under SSP2-4.5”). Generic “sea levels will rise” is AO1 only; numerical projection and its socio-economic consequences are AO2 application.

9. Coastal Management

Hard Engineering

Hard engineering uses large, built structures to resist wave energy and hold the coastline in place.

Approach	Description	Pros	Cons
Sea wall	Concrete / masonry wall at cliff or beach toe; often recurved to deflect waves	High protection; multi-functional (promenade)	£5,000–10,000 per m; reflects energy causing scour; ugly; terminal scour downdrift
Groynes	Timber or rock fences perpendicular to shore, trapping longshore drift	Builds up beach → dissipates wave energy; low cost (£10k–20k each)	Starves downdrift beaches (“terminal groyne syndrome” at Mappleton)

Rock armour (rip-rap)	Large boulders (typically granite, 3–10 t) placed at cliff toe	Absorbs wave energy via voids; relatively cheap; easy to recycle	Unightly; hazardous to walkers; boulders must be imported
Gabions	Wire cages of smaller rocks stacked at cliff base	Cheap; permeable, absorbs energy	Short lifespan (5–10 yr) — wire rusts
Revetments	Sloping timber or concrete structure to break wave energy before cliff	Cheaper than a sea wall; effective in sheltered settings	Requires regular maintenance; visually intrusive
Offshore breakwaters	Rock barriers placed offshore to induce wave breaking before they reach the coast	Reduce wave energy → encourage beach deposition	High cost; may interrupt sediment cell dynamics

Soft Engineering

Soft engineering works with natural processes to manage rather than resist the coast.

Approach	Description	Pros	Cons
Beach nourishment	Import sand/shingle (often dredged offshore) onto eroding beaches	Maintains natural appearance; dissipates wave energy; tourism value	Ongoing cost (£3,000–200,000 per 100 m per year); new material soon redistributed
Dune regeneration	Planting of marram grass; fencing to trap wind-blown sand	Low cost; enhances habitat; effective dissipator of storm surge energy	Needs protection from trampling; slow-developing
Managed realignment	Deliberate breach of an old defence line to let the sea reclaim low-lying land and create salt marsh	Sustainable; creates habitat; often cheaper long-term; reduces wave energy reaching remaining defences	Loss of farmland or property; compensation needed; politically controversial

Cliff stabilisation	Drainage, vegetation planting, regrading of slope angle	Addresses cliff-collapse cause (groundwater) directly	Slow results; does not stop marine erosion at cliff toe
Marsh / mangrove creation	Restoration or planting of intertidal vegetation	High biodiversity; carbon sequestration; wave attenuation	Slow to establish; requires suitable tidal range

Integrated Coastal Zone Management (ICZM)

ICZM: A holistic management approach that operates at the spatial scale of the entire sediment cell and the temporal scale of decades to a century. It combines physical processes, socio-economic needs (tourism, fishing, housing), ecological values and land-use planning through multi-stakeholder consultation. ICZM is adaptive — policies are regularly reviewed against monitored outcomes.

Shoreline Management Plans (SMPs)

SMP (Shoreline Management Plan): A non-statutory policy document, produced at sediment-cell level by the Environment Agency and local authorities, setting out the long-term management policy for each section of the England and Wales coast over three epochs (0–20, 20–50, 50–100 years). Each section is assigned one of four policy options.

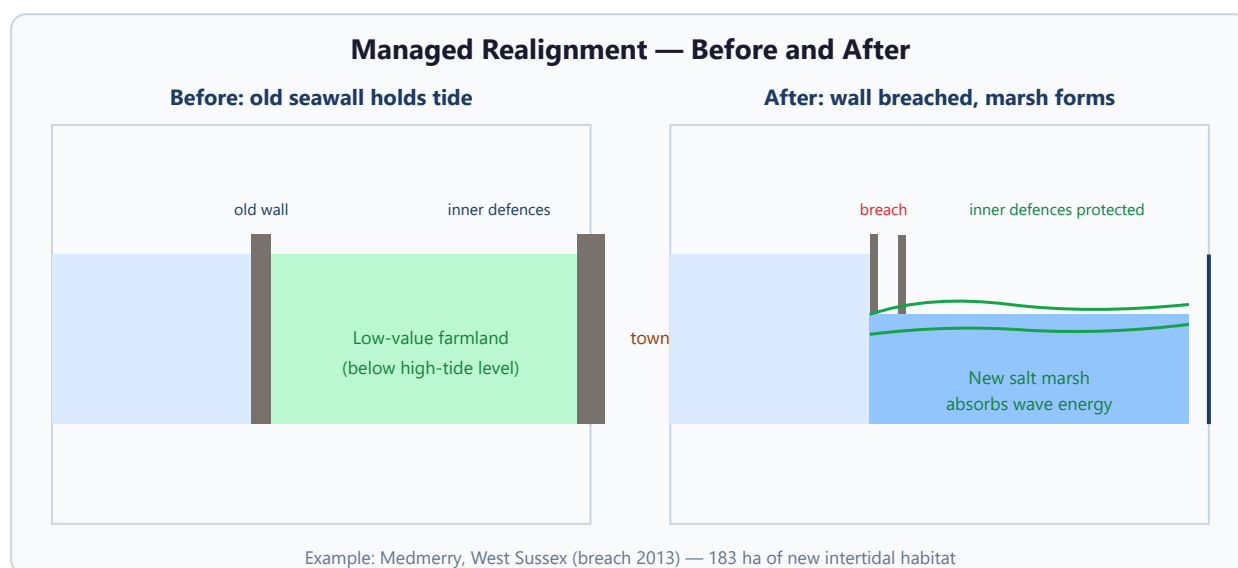
SMP Policy	Meaning	Typical application
Hold the Line (HTL)	Maintain the existing coastline using whatever engineering is required	High-value urban / infrastructure areas (e.g. central Bournemouth)
Advance the Line (ATL)	Build new defences seaward of the existing coastline	Rare in UK today; e.g. historic land reclamation at parts of the Humber
Managed Realignment (MR)	Allow the coast to move landwards under control, creating intertidal habitat	Low-value farmland backed by high-value assets (e.g. Medmerry, Wallasea)

No Active Intervention (NAI)

Permit natural processes (usually erosion) to continue unmanaged

Undeveloped cliffs of modest economic value (parts of Holderness)

Managed realignment: An engineered breach of an existing defence (usually an old sea wall) in order to let the sea flood low-lying land behind it. The newly flooded area reverts to salt marsh or mudflat, which absorbs wave energy and protects remaining inland assets. It is a flagship example of *working with* rather than *against* coastal processes.



Exam tip — SMP vocabulary: Use the abbreviations HTL, MR, NAI and ATL in answers on management. Eduqas mark schemes explicitly credit accurate use of SMP policy language. Remember that a single SMP typically contains *all four* policies in different sections of the coast — SMPs are not monolithic.

10. Case Studies

CASE STUDY 1 — Holderness Coast, East Yorkshire (UK)

Location: East Yorkshire, North Sea coast from Flamborough Head (chalk) in the north to Spurn Point (sand/shingle spit) in the south; ~61 km of rapidly eroding soft-rock coast.

Key facts: The fastest-eroding coast in Europe, averaging $1.8\text{--}2.0\text{ m yr}^{-1}$ and locally up to 10 m in a single storm. Cliffs are Quaternary-age boulder clay (till), a weakly cemented mix of sand, silt, clay

and pebbles dumped by the Devensian ice sheet. High wave energy: dominant north-easterly storm winds blowing across a long North Sea fetch produce powerful destructive waves; storm surges regularly overtop cliffs.

Processes: Hydraulic action and abrasion at the cliff toe combined with slumping of the clay during wet winters (pore-water pressure + lubrication). Longshore drift runs strongly from north to south, reaching Spurn Point where sediment accumulates. ~ 2 million $\text{m}^3 \text{yr}^{-1}$ of sediment feeds the Humber Estuary and southern North Sea budget.

Issues & conflicts: Around 30 villages (often quoted as 29) have been lost since Roman times (recorded list includes Ravenser Odd, Sunthorpe, Auburn). Present settlements at risk: Skipsea, Ulrome, Aldbrough, Cowden (caravan parks cliff-edge); gas terminal at Easington critical to UK supply. The B1242 coast road has been realigned several times. *Terminal groyne syndrome* at Mappleton: 1991 two rock groynes and revetment built to protect Mappleton (£2 million); beach held, but the village of Great Cowden 3 km south suffered accelerated erosion as drift was intercepted.

Management: Current SMP (Flamborough to Gibraltar Point, 2nd-generation) mostly specifies *No Active Intervention* given low cost-benefit ratios for most farmland, but *Hold the Line* at Bridlington, Hornsea, Withernsea, Mappleton, Easington gas terminal. Localised hard engineering (sea walls, rock armour, groynes) reflects high-value asset protection.

Evaluation: Illustrates classic management dilemma — protecting one community pushes erosion onto neighbours. Offers strong AO3 material on winners (Mappleton residents), losers (Great Cowden), sediment-budget disruption (reduced feed to Spurn Point, which periodically breaches) and the limits of hard engineering in a rapidly eroding soft-rock system.

CASE STUDY 2 — Medmerry Managed Realignment, West Sussex (UK)

Location: 2 km of low-lying coast west of Selsey, West Sussex, between Bracklesham and Selsey.

Background: The existing defence, a 3 km shingle bank, was regularly breached (most recently in 2008, causing over £5 million of damage). Behind it lay a large caravan park, a sewage works, low-grade farmland and the town of Selsey. Re-profiling the bank with bulldozers each winter was costing the Environment Agency up to £300,000 a year, with declining effectiveness.

Scheme: The UK's largest-ever open-coast managed realignment, delivered by the Environment Agency 2011–2013 at a total cost of ~£28 million. The old shingle bank was breached in one 110 m

gap in September 2013; new inland embankments (7 km of them, up to 6 m high) were built to retain the sea within the designated area. The 183 ha of newly intertidal land rapidly developed into mudflat and pioneer salt marsh.

Outcomes:

- Selsey, 348 homes and the B2145 road protected to at least a 1-in-1000-year flood standard.
- New intertidal habitat designated an RSPB nature reserve; dramatic colonisation by breeding avocets, little terns, lapwings and redshanks within 3 years.
- Benefit-cost ratio calculated at ~£90 million of flood damage avoided over 100 years vs. £28 million construction.
- Creates natural wave buffer — the new marsh attenuates storm surge reaching the inland embankment.

Issues: Loss of working farmland; compensation paid to landowners. Some local opposition to “abandonment” mindset; perception that government was retreating rather than defending.

Evaluation: A flagship example of soft engineering / MR, used by the Environment Agency to demonstrate that working with natural processes can deliver better protection at lower whole-life cost than indefinite maintenance of hard engineering. Illustrates how SMP policy can shift from HTL to MR as cost-benefit changes over time.

CASE STUDY 3 — Sundarbans, Bangladesh / India

Location: The world's largest continuous mangrove forest (~10,000 km²), straddling the Ganges–Brahmaputra–Meghna delta across south-west Bangladesh and West Bengal, India. Population of the wider delta region ~4.5 million in the mangrove fringe, >100 million in the broader delta.

Key facts: Low-lying (much <2 m elevation above mean sea level), acutely exposed to the Bay of Bengal which channels tropical cyclones (“northern cyclones”) producing 5–10 m storm surges. Delta sediment supply has been reduced by upstream dam construction (Farakka Barrage, 1975), while subsidence of the delta and eustatic sea-level rise give a combined relative sea-level rise of 5–8 mm yr⁻¹, several times the global mean.

Issues:

- Permanent inundation of low-lying islands: Lohachara (submerged 1996) and New Moore Island (2010) already lost.

- Saltwater intrusion into groundwater and paddy fields reducing rice and fish yields.
- Cyclone Sidr (2007) killed ~3,400 and displaced ~2 million; Cyclone Amphan (2020) caused roughly US\$13 billion of damage, mostly in India.
- Climate migration: projected 13–26 million internal displaced persons in Bangladesh by 2050.

Management: Limited budget rules out UK-style hard engineering. Responses include:

- Polder embankment system (139 coastal polders, ~5,700 km of embankment built since the 1960s), with a World Bank-supported US\$400 m upgrade programme (CEIP-I).
- Large-scale mangrove planting on newly accreted coastal land by the Bangladesh Forest Department since the 1960s; mangrove belts substantially reduce wave height and storm-surge energy.
- Elevated cyclone shelters (thousands built), early-warning systems via mobile phones.
- Planned relocation of vulnerable communities and “climate-resilient” agriculture (floating vegetable gardens, salt-tolerant rice varieties).

Evaluation: Starkly illustrates the asymmetry of climate vulnerability — Bangladesh contributes <0.5 % of global emissions but faces among the highest sea-level rise impacts. Mangrove-based “blue infrastructure” is effective but threatened by its own loss to salinisation and shrimp-farm conversion. Powerful AO3 material on sustainability, equity and the limits of local adaptation when the primary driver is global.

CASE STUDY 4 — Jurassic Coast UNESCO WHS, Dorset / East Devon (UK)

Location: 155 km of coast from Orcombe Point (Exmouth, Devon) in the west to Old Harry Rocks (Studland, Dorset) in the east; England's first natural UNESCO World Heritage Site (2001).

Significance: Provides a continuous “walk through time” of 185 Ma of Earth history (Triassic red sandstones in the west through to Cretaceous chalk in the east). The interaction of varying rock types, structure and Pleistocene sea-level change has produced a textbook sequence of landforms:

- **Lulworth Cove** — a near-perfect almond-shaped cove in a concordant coast. The resistant Portland Limestone outer wall was breached during post-glacial sea-level rise; the softer Wealden clays behind were rapidly eroded, then the breaching widened until stopped by the resistant Chalk inner wall.

- **Stair Hole** — an earlier stage of the same process: a narrow entrance through the Portland Limestone, with dramatic folding of the Purbeck beds visible in the cliff.
- **Durdle Door** — a natural arch in vertical Portland Limestone, formed by wave erosion of a line of weakness; its stack ("Bull") marks where an earlier arch has collapsed.
- **Chesil Beach** — 29 km tombolo of shingle (pea-sized in the west, fist-sized at Portland), enclosing the Fleet lagoon. Probably a Devensian relict feature driven onshore by post-glacial sea-level rise.
- **Swanage Bay and Ballard Down** — discordant coast: chalk headlands (Ballard Point, with Old Harry Rocks stacks and stumps) flanking a bay cut in softer Wealden sands.

Management: A mosaic of SMP policies: *Hold the Line* at Swanage seafront, Weymouth and Lyme Regis; *No Active Intervention* over most undeveloped cliffs to preserve the WHS's designating geological exposures. Lyme Regis has received ~£40 million of phased beach nourishment, offshore breakwaters and cliff stabilisation since 1990 ("Lyme Regis Environmental Improvements").

Evaluation: Demonstrates (a) the primacy of geology in controlling landform style, (b) the interaction of rock structure with Pleistocene sea-level rise, and (c) the tension between protecting UNESCO designation and protecting the towns of residents. Excellent case study for 15-mark extended responses integrating process, landform and management.

Case Study	Focus	Key process	Key management
Holderness	Rapid erosion of soft-rock coast	Cliff retreat, LSD, slumping	Mixed SMP (HTL + NAI); terminal groyne conflict
Medmerry	Managed realignment	Marsh accretion, wave attenuation	Soft engineering — MR
Sundarbans	SLR vulnerability & adaptation	Cyclone surge, salinisation, subsidence	Mangrove planting, polders, migration
Jurassic Coast	Geology & landform diversity	Differential erosion on discordant & concordant coasts	Mixed SMP; nourishment at Lyme Regis

11. Evaluating Coastal Management

A defining feature of top-band Eduqas answers is evaluation (AO3). Management is never neutral: it produces *winners* and *losers*, and its sustainability must be judged on three dimensions.

- **Environmental sustainability** — Hard defences often disrupt sediment cells, starve downdrift beaches and produce coastal squeeze that eliminates intertidal habitat. Soft engineering (dune regeneration, marsh creation, MR) tends to enhance biodiversity and sequester carbon.
- **Economic sustainability** — Whole-life costs matter. A sea wall costing £5,000 per metre has 50–100-year lifetime and high maintenance; beach nourishment is cheaper upfront but ongoing. Benefit-cost ratios favour HTL only where asset values are high (central Bournemouth) and MR where they are low (agricultural land with flooded hinterland).
- **Social sustainability** — Public acceptance varies: residents of Happisburgh (Norfolk) campaigning for hard defences under NAI policy; farmers displaced by Medmerry initially resistant but compensated and eventually supportive. Equity is central — Bangladesh's climate-vulnerable poor have limited alternatives compared with Essex farmers.

Climate change now alters the long-term cost-benefit of every policy. Hold the Line schemes assume a static sea level; with 0.5–1 m of projected rise, they will need to be heightened, rebuilt or abandoned. This is the principal AO3 argument in favour of MR and ICZM: *adaptation rather than resistance*.

Exam tip — structuring 15-mark evaluative essays: A typical Eduqas 15-mark stem is “Assess the effectiveness of coastal management approaches in maintaining a coastline”. Use a PEEL × 4 structure: (1) Hard engineering — strengths + named example (e.g. Mableton groynes); (2) Hard engineering — costs + sediment-cell disruption; (3) Soft engineering / MR — Medmerry success; (4) ICZM / holistic — sustainability under climate change. Finish with a clear overall judgement anchored in “it depends on spatial scale, time horizon and value of assets”.

Exam tip — integrating process, landform and case study: Examiners complain that candidates describe landforms OR processes OR case studies in isolation. In a 15-mark answer, integrate all three: start with the process (e.g. hydraulic action in jointed rock), link to the landform sequence (crack → cave → arch → stack), and then anchor with a named case study (Old Harry Rocks). One integrated paragraph beats three disconnected ones.

Exam tip — A2 vs GCSE depth: At A-level, named examples must be accompanied by quantified data (retreat rates in m yr^{-1} , costs in £ millions, percentages). “The Holderness coast erodes very fast” is GCSE-level. “The Holderness coast erodes at $\sim 2 \text{ m yr}^{-1}$, supplying $\sim 2 \times 10^6 \text{ m}^3 \text{ yr}^{-1}$ of sediment to the southern North Sea cell” is A2.

Exam tip — case-study balance: A 15-mark answer rarely requires all four case studies. Pick two that contrast — e.g. Holderness (erosion + HTL/NAI mix) and Medmerry (MR) — and go deep. Names of settlements, quantified data, dates and outcomes beat superficial coverage of four examples.

Exam tip — diagram inclusion: A well-labelled sketch is a quick way to show AO1 understanding in landform questions. A clear labelled sketch of cliff recession, stack formation or spit development (with named components: notch, overhang, recurve, salt marsh behind) can secure AO1 marks quickly. Practise these under timed conditions — 2–3 minutes per diagram.

Exam tip — climate-change framing: Any 15-mark coast question can usefully end with a climate-change paragraph: IPCC AR6 projections, rising frequency of destructive storms, coastal squeeze, and implications for sustainability of management. This is high-value AO3 and signals synoptic awareness.

12. Additional Depth — Sediment Cells of England and Wales

The 11 sediment cells of England and Wales form the spatial backbone of SMP planning. Candidates are expected to locate their chosen case-study area within a named cell and comment on how its sediment budget has been altered by human action.

Cell	Boundaries	Notable features
1	St Abb's Head → Flamborough Head	Northumberland coast; dunes at Lindisfarne; spits at Budle Bay

2	Flamborough Head → The Wash	Holderness — dominant erosion source; Spurn Point sink
3	The Wash → North Foreland (Thames Estuary)	Low-lying East Anglia; managed realignment at Wallasea Island
4	North Foreland → Selsey Bill	Estuary-dominated sediment stores
5	Selsey Bill → Portland Bill	Chesil Beach tombolo; Jurassic Coast WHS
6	Portland Bill → Land's End	Rias (Fal, Dart); high-energy Atlantic-facing coast
7	Land's End → St David's Head	North Cornwall / Devon; surfing beaches; dune complexes
8	St David's Head → Great Orme	Cardigan Bay; raised beaches on Llŷn Peninsula
9	Great Orme → Solway Firth	Liverpool Bay; Sefton dune complex; Dee Estuary marshes
10–11	Scottish border to western Solway	Limited sediment exchange; many raised beaches

Sub-cells divide each major cell into management units. Sediment rarely crosses cell boundaries because large headlands (Flamborough, Land's End, Portland Bill) either trap drift or deflect it offshore. This self-contained nature means that *management in one cell does not usually propagate damage to another*, which is why the sediment cell is used as the SMP planning unit.

13. Applying the Sediment Budget Concept

A sediment budget is a mass-balance calculation for a cell or sub-cell. For a simplified beach compartment:

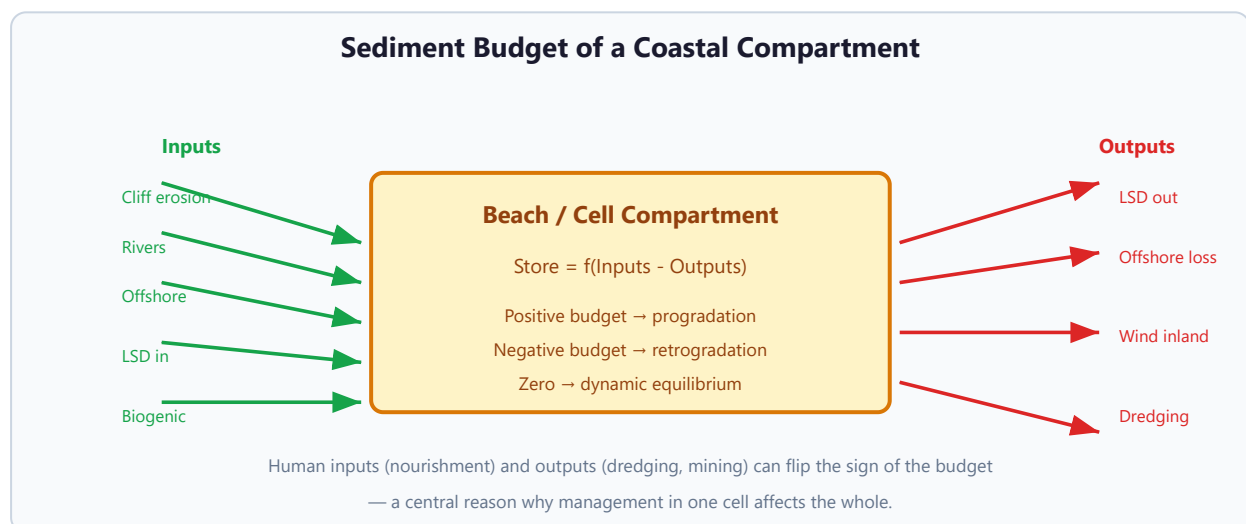
$$\Delta\text{Storage} = \sum\text{Inputs} - \sum\text{Outputs}$$

Worked example — a 5 km stretch of Holderness coast between Barmston and Hornsea:

- **Inputs:** cliff erosion (2 m yr^{-1} retreat $\times 30 \text{ m high cliff} \times 5,000 \text{ m} \times 0.6$ sand fraction) $\approx 180,000 \text{ m}^3 \text{ yr}^{-1}$ of sand-grade sediment; upcoast drift in $\approx 50,000 \text{ m}^3 \text{ yr}^{-1}$.
- **Outputs:** downcoast drift out $\approx 220,000 \text{ m}^3 \text{ yr}^{-1}$; offshore loss $\approx 15,000 \text{ m}^3 \text{ yr}^{-1}$.
- **Balance:** $230,000 - 235,000 = -5,000 \text{ m}^3 \text{ yr}^{-1} \Rightarrow$ slight net loss, consistent with observed narrowing of mid-Holderness beaches.

Intervention at Mappleton (groynes trapping $\sim 80,000 \text{ m}^3 \text{ yr}^{-1}$ of drift) converts the Great Cowden compartment below to a strongly negative budget, explaining the accelerated erosion. This is the kind of quantified reasoning that secures full AO2 marks.

13a. Sediment Budget — Visual



14. Coastal Flooding and Storm Surges

The Eduqas spec also requires students to understand **coastal flooding** and its interaction with sea-level rise. Flooding occurs when water levels exceed defence crests, which can result from:

- **High spring tides** — predictable; maximum at full and new moon.
- **Low atmospheric pressure** — each 1 hPa fall raises sea level by $\sim 1 \text{ cm}$ (inverse barometer).
- **Wind setup** — sustained onshore wind piles water against a coast; in a shallow basin like the southern North Sea this can add 2–3 m.
- **Wave setup and runup** — additional water-level elevation in the surf zone.

UK case — 1953 East Coast Flood: A surge of up to ~3 m (2.97 m at King's Lynn) on top of a spring tide flooded ~57,000 ha (140,000 acres) in eastern England, killed 307 and damaged 24,000 properties. This disaster led to the Thames Barrier (1982) and the modern Environment Agency flood-warning network.

2013 “Xaver” surge: Exceeded 1953 levels at some gauges but caused far less damage owing to improved defences and warning — evidence that effective management reduces vulnerability even as hazard magnitude rises.

Exam tip — named storm events: Eduqas questions frequently ask for a named storm or flood event in the UK or overseas (e.g. Cyclone Sidr 2007, Storm Xaver 2013, Hurricane Katrina 2005). Have dates, wind speeds, surge heights, death tolls and economic cost at your fingertips.

15. Human Activity and the Coastal System

Humans act as both inputs and outputs to the coastal system and can override many natural controls:

- **Dam construction** on rivers traps sediment; Nile delta erosion has accelerated since the Aswan High Dam (1970) cut sediment delivery by ~98 %.
- **Beach mining** for sand (construction) or heavy minerals has stripped beaches in Morocco, India and parts of the Caribbean, triggering positive-feedback cliff retreat.
- **Coastal land claim** (Netherlands, Singapore, Hong Kong) extends coasts seaward but permanently alters the sediment cell.
- **Pollution** — nutrient runoff eutrophies estuaries; microplastic inputs now detected in every major coast.
- **Tourism infrastructure** — promenades, marinas and groynes that were built for amenity or navigation reshape drift patterns.

The concept of the *Anthropocene coast* — where human activity is a first-order control on sediment budgets and shoreline position — is an increasingly prominent framing at A-level and usefully integrates physical and human geography.

16. Synoptic Summary

Coastal landscapes (Eduqas 1.1) bring together physical, environmental and human geography. Above all, the spec rewards candidates who treat the coast as a *system* in *dynamic equilibrium*, who explain *processes* before *landforms*, who support argument with *named quantified case studies*, and who *evaluate* management in terms of environmental, economic and social sustainability in a changing climate. Mastery of the four case studies above (Holderness, Medmerry, Sundarbans, Jurassic Coast) and fluency with SMP policy vocabulary (HTL, MR, NAI, ATL) provides the backbone for answering the majority of 5-, 8- and 15-mark questions on this topic.

Key revision checklist:

- Can you define and apply *dynamic equilibrium*, *feedback*, *sediment cell*, *sediment budget*?
- Can you draw and label cliff recession, stack formation, a spit, a sand-dune psammosere and a managed realignment cross-section?
- Can you explain one erosional, one depositional, one emergent and one submergent landform with a named UK example?
- Can you compare the four SMP policy options with named locations for each?
- Can you quote 3–4 quantified facts per case study?
- Can you evaluate a management scheme on environmental, economic and social sustainability?
- Can you integrate IPCC AR6 sea-level data into a climate-change paragraph?